

Common Admission Test (CAT 2024)

Official Detailed Syllabus & Blueprint: Physics (Test Code 101)

Curriculum: NCERT Class XI & XII Standard

Level: Undergraduate & 5-Year Integrated M.Sc

Session: 2024

Parameter	Examination Specification
Examination Mode	Computer Based Test (CBT) Online Examination
Total Questions (Physics Section)	75 Multiple Choice Questions (MCQs)
Marking Scheme	+3 marks for each correct response; -1 mark for each incorrect response (Negative marking applies)
Maximum Marks (Physics)	225 Marks
Eligible Programs	B.Tech, 5-Year Integrated M.Sc in Physics, Chemistry, Mathematics, Statistics, Marine Biology

Class XI Syllabus Modules

Unit 1: Physics and Measurement

- Scope & Dimensions:** Units of measurements, System of Units (SI Units), fundamental and derived units. Dimensions of physical quantities, dimensional analysis and its applications. Least count, significant figures, errors in measurement and combination of errors.

Unit 2: Kinematics

- Motion in 1D & 2D:** Frame of reference, Motion in a straight line: Position-time graph, speed and velocity. Uniform and non-uniform motion, average speed and instantaneous velocity, uniformly accelerated motion, velocity-time and position-time graphs, kinematic equations for acceleration. Scalars and vectors, vector addition, resolution, dot product and cross product. Unit vectors. Projectile motion and uniform circular motion.

Unit 3: Laws of Motion

- **Dynamics:** Force and inertia, Newton's first, second, and third laws of motion. Momentum, impulse and law of conservation of linear momentum with applications. Equilibrium of concurrent forces. Static and kinetic friction, laws of friction, rolling friction, lubrication. Dynamics of uniform circular motion: Centripetal force, banking of roads.

Unit 4: Work, Energy and Power

- **Energy & Collisions:** Work done by a constant force and variable force; Kinetic energy and work-energy theorem. Power. Potential energy of a spring, conservative and non-conservative forces, conservation of mechanical energy. Elastic and inelastic collisions in one and two dimensions.

Unit 5: Rotational Motion & System of Particles

- **Rigid Body Dynamics:** Centre of mass of a two-particle system, momentum conservation and centre of mass motion. Torque, angular momentum and its conservation with applications. Moment of inertia, radius of gyration, parallel and perpendicular axes theorems. Moments of inertia of simple geometrical bodies. Equilibrium of rigid bodies.

Unit 6: Gravitation

- **Planetary Motion & Fields:** Universal law of gravitation. Acceleration due to gravity and its variation with altitude and depth. Kepler's laws of planetary motion. Gravitational potential energy and gravitational potential. Escape speed, orbital velocity of satellites, geostationary satellites.

Unit 7: Properties of Bulk Matter & Fluids

- **Elasticity & Hydrodynamics:** Hooke's law, stress-strain relationship, Young's modulus, bulk modulus, modulus of rigidity. Pascal's law and its applications (hydraulic lift). Viscosity, Stokes' law, terminal velocity, streamline and turbulent flow, critical velocity, Reynolds number. Bernoulli's principle. Surface energy, surface tension, angle of contact, excess pressure, capillary rise.

Unit 8: Thermodynamics & Kinetic Theory of Gases

- **Thermal Physics:** Thermal equilibrium, zeroth, first and second laws of thermodynamics. Isothermal, adiabatic, isobaric, and isochoric processes. Reversible and irreversible processes, Carnot engine and efficiency. Kinetic theory of gases: Assumptions, concept of pressure, kinetic energy and temperature, RMS speed, degrees of freedom, law of equipartition of energy, specific heats of gases, mean free path.

Unit 9: Oscillations and Waves

- **SHM & Acoustics:** Periodic motion, period, frequency, simple harmonic motion (SHM) and its equation; Phase, oscillations of a spring and simple pendulum. Free, forced, and damped oscillations, resonance. Wave motion: Longitudinal and transverse waves, speed of wave motion, displacement relation for a progressive wave. Principle of superposition of waves, reflection of waves, standing waves in strings and organ pipes, fundamental mode and harmonics, beats, Doppler effect in sound.

Class XII Syllabus Modules

Unit 10: Electrostatics

- **Charges & Fields:** Electric charges, Coulomb's law, superposition principle, continuous charge distribution. Electric field, electric field lines, electric dipole, torque on a dipole. Gauss's law and applications (wire, sheet, sphere). Electric potential, potential difference, equipotential surfaces, electrical potential energy. Conductors and insulators, dielectrics, capacitors and capacitance, combination of capacitors, energy stored in a capacitor.

Unit 11: Current Electricity

- **DC Circuits:** Electric current, drift velocity, mobility, Ohm's law, V-I characteristics, electrical resistance, resistivity and conductivity. Temperature dependence of resistance. Internal resistance of a cell, terminal voltage, emf, cells in series and parallel. Kirchhoff's laws and applications: Wheatstone bridge, Metre bridge, Potentiometer (principle, comparison of emf, internal resistance).

Unit 12: Magnetic Effects of Current and Magnetism

- **Biot-Savart & Ampere Laws:** Biot-Savart law, magnetic field due to current loop and solenoid. Ampere's circuital law. Force on a moving charge in magnetic and electric fields (Lorentz force), cyclotron. Force on a current-carrying conductor in a uniform magnetic field. Force between two parallel currents, torque on current loop, moving coil galvanometer and conversion to ammeter/voltmeter. Magnetic dipole, bar magnet, Earth's magnetic field and magnetic elements. Para-, dia-, and ferromagnetic substances, hysteresis.

Unit 13: Electromagnetic Induction & Alternating Currents

- **Faraday & AC Circuits:** Electromagnetic induction, Faraday's laws, induced EMF and current, Lenz's law, Eddy currents. Self and mutual inductance. Alternating currents, peak and RMS value of AC; reactance and impedance; LCR series circuit, resonance, power in AC circuits, wattless current. AC generator and transformer.

Unit 14: Electromagnetic Waves

- **EM Spectrum:** Displacement current, characteristics of EM waves, transverse nature. Electromagnetic spectrum (radio, microwaves, infrared, visible, ultraviolet, X-rays, gamma rays), properties and applications.

Unit 15: Optics (Ray & Wave Optics)

- **Ray Optics:** Reflection of light, spherical mirrors, mirror formula. Refraction, total internal reflection, optical fibres, refraction at spherical surfaces, thin lens formula, lens maker's formula, magnification, power of a lens, combination of thin lenses. Dispersion through a prism. Optical instruments: Microscopes and astronomical telescopes (refracting and reflecting).
- **Wave Optics:** Wave front and Huygens' principle, reflection and refraction of plane wave. Interference, Young's double-slit experiment, fringe width. Diffraction due to a single slit, width of central maximum. Polarization, Brewster's law, uses of plane polarized light and Polaroid.

Unit 16: Modern Physics & Electronic Devices

- **Quantum, Nuclear & Electronics:** Photoelectric effect, Hertz and Lenard observations, Einstein's photoelectric equation, de Broglie relation, Davisson-Germer experiment. Alpha-particle scattering, Rutherford model, Bohr model, energy levels, hydrogen spectrum. Composition and size of nucleus, atomic masses, mass-energy relation, mass defect, binding energy, nuclear fission and fusion. Energy bands in conductors, semiconductors and insulators; intrinsic and extrinsic semiconductors, p-n junction, forward and reverse bias, diode as rectifier, I-V characteristics of LED, photodiode, solar cell, Zener diode as voltage regulator. Logic gates (OR, AND, NOT, NAND, NOR).